Discussion: what the evidence supports

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The 9 studies probe distinct parts of one categorical, factor-based framework rather than repeating a single benchmark. Their common result is narrow but useful: the client construction contains its standard-Bayes limit and the server has a named project log-linear-pool recovery corner, and behavior away from those limits can be measured under declared contamination, sampling, and model assumptions. The recovery identities are formal; the performance results are conditional simulation evidence.

The recovery limit is the formal anchor I

The coherence of the framework rests on the KL/NLL client limits and the zero-robustness project-pool identity (eq. 7, eq. 6). This is an identity of the stated categorical implementation, not an asymptotic claim about every generalized Bayesian model or a reconstruction of the complete source protocol. The server aggregator returns the log-linear pool (eq. 6) with maximum deviation 0; the generalized posterior returns the closed-form Bayes update with deviation 5.55e-17; and the Rényi divergence, β-loss, and robust cross-entropy recover their KL/NLL limits with residuals 0, 0, and 0. Theorem 5, Corollary 6, Lemma 3, and Proposition 4 state the formal limits; the recovery checks in sec. 7.1 are the executable falsification harness.

What the study suite jointly shows I

Read together, the suite separates into two kinds of result: identities that hold exactly at the standard corner, and performance contrasts that are explicitly conditional on the operating regime. The distinction is the point of the joint reading — it says which claims travel and which are tied to the declared world.

Studies 1–3 sit at the standard corner. In the reduced categorical source-mechanism analogue protocol, belief sharing lowers mean variational free energy by $\Delta\bar{F} = 3.3109$ nats (sec. 7.2, fig. 8); Dirichlet language learning reduces KL from 3.4231 to 0.0027 nats (sec. 7.3, fig. 10); and Bayesian model reduction accepts the redundant-pruning candidate while rejecting the supported-column control (sec. 7.4, fig. 11). These are mechanistic checks, not evidence that the implementation matches every detail of the source simulations.

What the study suite jointly shows I

Study 4 steps away from the corner by holding the hidden state and attack target fixed while redrawing matched contaminated colonies. The standard pool degrades across the declared rate grid. The server-side heuristic is not uniformly better: its robust members are similar to or slightly below the standard pool at low contamination, then the selected pooled display member separates in its favor under severe contamination. At the largest swept rate, AR reaches 0.9880 against 0.6928 for the standard pool (tbl. 3; the sweep figure fig. 12 plots the separate matched-trial gallery colony, whose largest-rate separation is smaller). This regime dependence is a result, not a nuisance to be hidden: robustness can cost efficiency when the attack is weak and pay off when the declared contamination is severe.

The structural extension studies (Studies 5–9) sharpen the conditional half of that split rather than adding further corner checks, and they are reported with their negative contrasts intact. Communication is not uniformly beneficial: the cross-study summary (fig. 27) reports each study's federation benefit in its native units with seed-level intervals — several studies clearly positive, the moving-world EFE and two-level hierarchical studies approximately zero — so pooling helps when views are complementary and can be unnecessary or mildly costly when the agents already agree. Adding hierarchical depth is held to the same standard — the two-level stack does not beat the flat baseline on location accuracy (the paired gap is a small, statistically reliable cost) (sec. 13.9), earning its place only by additionally resolving the context latent above chance. The joint lesson is therefore not that federation or depth is always worth its cost, but

What the study suite jointly shows I

that the suite measures the regimes in which each one is.

What this simulation identifies—and what it does not I

The primary robustness estimand is the matched-trial mean difference in consensus accuracy conditional on the seeded true state and attack geometry. The 960 paired trials quantify Monte Carlo variation for that estimand; they do not average over hidden states, attack targets, adaptive adversaries, or real deployments. The cross-study layer reduces its matched trials within seed before seed-level summaries, so clients and within-seed trials are not silently counted as independent replicates. This follows simulation study guidance to declare the estimand and Monte Carlo unit explicitly (Morris et al. 2019; Koehler et al. 2009), and the bootstrap interval follows the declared resampling unit rather than treating nested observations as a flat sample (Loy and Korobova 2021).

What this simulation identifies—and what it does not I

The consequence is a more informative claim boundary. The sweep supports a conditional statement about this contamination mechanism and these categorical beliefs; it does not establish universal Byzantine tolerance, calibration, or truth recovery. The result also does not identify a single universally best robustness parameter independent of the operating regime: the standard pool is preferable at the low-contamination cells in this run, while at least one robust member has a BH-rejected positive contrast in the declared high-contamination verdict; the tied display set and deterministic tie-break are reported separately. Those are precisely the conditions a future adaptive or deployment study must vary.

The robustness verdict is conditional and statistically qualified I

The headline comparison is computed by the statistics module (sec. 7.5.1), not typed into the prose. Across 960 matched trial replicates nested within the fixed seeded world at contamination rate 0.800, each robust server member is compared with the standard pool by a Wilcoxon signed-rank test (Wilcoxon 1945) and the declared family of p-values is adjusted by Benjamini–Hochberg FDR (Benjamini and Hochberg 1995). A method wins only when the adjusted null is rejected and its effect is positive, here at $q = 1.11 \times 10^{-158}$ with effect size 1.0000 (large). The observed-effect power and prospective sample-size calculation are planning quantities, not independent confirmation of the result.

The robustness verdict is conditional and statistically qualified I

The paired design is appropriate for the controlled comparison because each condition shares the seed, true state, and attack geometry. Its interpretation remains limited: the signed-rank test concerns the distribution of paired differences, not equality of raw means, and BH controls expected false-discovery proportion within the declared family rather than the probability of any false positive. The confidence intervals are percentile bootstrap intervals over the matched-trial unit. Seed-level review-grid results are reported separately and do not treat the primary trial count as a population of independent worlds. These qualifications make the verdict narrower, but also make it reproducible and falsifiable.

The unifying narrative would be dishonest if it let the aggregation heuristic inherit the guarantees of the client update or variational server objective. The **client-side** β/rcce generalized-Bayes update is the FedGVI-faithful axis (Mildner et al. 2025): it is derived from the generalized-Bayes objective (eq. 1), limits to NLL/Bayes as its loss parameter tends to zero, and carries the loss-specific bounded-influence result (eq. 4, eq. 5). The **server-side** `robust_aggregate` divergence-reweighting is a complementary heuristic whose proven property is the recovery limit only (eq. 7). The **variational server** axis is objective-backed and supplies a proven raw effective-weight bound, with conservative accuracy behavior. No figure, statistic, or sentence transfers a client guarantee to the heuristic or a variational weight bound to the heuristic's accuracy verdict; sec. 8.13 states the boundary in full.

Accuracy and effective-weight control can be traded explicitly I

The F_λ family (sec. 11.5) supplies empirical grid evidence that accuracy and effective-weight control need not be a fixed binary choice within the tested objective family. At $\lambda = 1.0$ the implementation recovers the current variational objective bit-for-bit; lower explored temperatures sharpen the consensus toward the heuristic while preserving the stated raw-weight bound under its assumptions. This is not a derivation of the heuristic from an objective. The open problem is to identify an objective whose minimizer is both competitive across contamination regimes and accompanied by a theorem that survives beyond the present categorical construction.

Why the boundary matters downstream I

The practical lesson is not that every robust rule should replace belief sharing. It is that a multi-agent active-inference system can expose separate controls for client updating, server reweighting, and objective-backed consensus, while retaining a tested route back to the ordinary pool. That separation tells a builder what can be promised: exact recovery at the named corner, conditional empirical behavior under the declared attack, and no silent transfer of a client-side or variational theorem to a different server heuristic. The result is a usable research contract for extending belief-sharing systems without confusing a simulation win with a general robustness guarantee.